Course Title	Deep Learning	L	T	P	C	Periods
Course Code	UCSH-DL	3	0	2	4	70
Course Category	DSE					

Course Objectives: Introduce the main deep learning architectures used in the industry. Develop a strong understanding of the training mechanisms. Expose students to Advanced topics like Attention based models, Deep Generative models

Course Outcomes: Through this course, students will be able to:

1. Develop deep learning solutions to any AI related problem
2. Apply different optimization techniques to train neural networks
3. Apply different dimensional reduction techniques using neural networks
4. Develop and train a CNN model to solve a real world problem based on Image data.
5. Develop and train a RNN model to solve a real world problem based on sequential data
6. Have a strong understanding of the internal workings of advanced deep learning architectures like VAEs and GANs
7. Implement several deep learning algorithms from scratch
8. Prototype solutions in Python

Prerequisites: Machine Learning, Optimization

Unit	Title	Contents	Periods
1	An Introduction to Neural Networks	McCulloch Pitts Neuron, Perceptrons, Perceptron Learning Algorithm and Convergence, Multilayer Perceptron (MLPs), Representation Power of MLPs, Sigmoid Neurons, Gradient Descent, FeedForward Neural Networks, Backpropagation.	8
2	Gradient Descent Strategies	Gradient Descent (GD), Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp and Adams	5
3	Embedding and Representation Learning	Review of Principal Component Analysis, Autoencoders and relation to PCA, Regularization in autoencoders, Denoising autoencoders, Sparse autoencoders, Contractive autoencoders	5
4	Regularization for Deep Neural Networks	Bias variance Tradeoff, $\square^{2}$- regularization, Early stopping, Dataset augmentation, Parameters sharing and tying, Injecting noise at input, Ensemble methods, Dropout, Greedy Layer Wise Pre-training, Better	6

		activation functions, Better weight initialization methods, Batch Normalization	
5	Convolutional Neural Networks	The basic structure of convolutional Network: Padding, Strides, ReLU Layer, Pooling, and Fully Connected Layers; Case studies of Convolutional Architectures: AlexNet, ZFNet, VGGNet, GoogLeNet, ResNet; Visualizing Convolutional Neural Networks, Guided Backpropagation.	6
6	Recurrent Neural Networks	Recurrent Neural Networks, Backpropagation through time (BPTT), Vanishing and Exploding Gradients, Truncated BPTT, GRU, LSTMs	6
7	Advanced topics in Deep Learning	Encoder Decoder Models, Attention Mechanism, Attention over images, Variational autoencoders, Generative Adversarial Networks (GANs)	6

Key Text(s):

1.Ian Goodfellow, YoshuaBengio, Aaron Courville, Deep Learning, The MIT Press, 2016
2.Nikhil Buduma, Nicholas Locascio, Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms, O'ReillyMedia, 2017
3. Charu C. Agarwal, Neural Networks and Deep Learning: A Textbook, Springer, 2018

References:

1.Duda, R.O., Hart, P.E., and Stork, D.G, Pattern Classification, Wiley-Interscience, Second Edition, 2001
2. Theodoridis, S. and Koutroumbas, K., Pattern Recognition. Fourth Edition, Academic Press, 2008
3. Russell, S. and Norvig, N. Artificial Intelligence: A Modern Approach, Prentice Hall Series in Artificial Intelligence. 2003
4. Bishop, C. M. Neural Networks for Pattern Recognition. Oxford University Press. 1995.
5. Hastie, T., Tibshirani, R. and Friedman, J. The Elements of Statistical Learning. Springer. 2001.
6. Koller, D. and Friedman, N. Probabilistic Graphical Models. MIT Press. 2009.

UCSH-DL: Practicals - Deep Learning

Unit	**Title**	**Contents**	**Periods**
1	PyTorch Basics	Why PyTorch?, Basics of Tensors: Tensors, indexing, size, offset, slicing, dtypes, moving tensors to GPU	10